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REPORT

Analysis of Water Quality Parameters in Delhi's River Water

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Hydrological Profiling of the Yamuna River: Field Research Report

Summary

This report details a field research study conducted in Delhi to assess the water quality and ecological health of the Yamuna River. The study identifies a catastrophic decline in water quality as the river flows through the national capital, primarily driven by systemic failures in waste management and industrial discharge.

Introduction and Significance

The Yamuna River, originating from the Yamunotri Glacier in Uttarakhand, is the longest tributary of the Ganga system. It serves as a vital resource for Delhi, yet it currently faces extreme environmental degradation.

- **Population Support:** Primary water source for 35.5 million people in the Delhi region.
 - **Ecological Role:** Critical for groundwater recharge, aquatic biodiversity, and local agriculture.
 - **Pollution Overview:** Despite representing only 2% of the river's total length, Delhi is responsible for approximately 80% of the Yamuna's total pollution load.
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Site-Wise Hydrological Analysis

The research team analysed water samples from several key locations:

- 1) **Ramghat**
- 2) **Confluence Point** (where the Najafgarh Drain meets the Yamuna)
- 3) **Industrial Water** (Near Nizamuddin Metro Station)
- 4) **Najafgarh Drain**
- 5) Tap Water Data from internet as a reference

- All the water samples were taken from the **marginal zone** of the sample collection sites.
 - All the tests were done after **two days** of the sample collection day.
 - The samples were stored in absence of sunlight at room temperature.
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PRIMARY WATER QUALITY CRITERIA FOR BATHING WATER
(Water used for organised outdoor bathing)

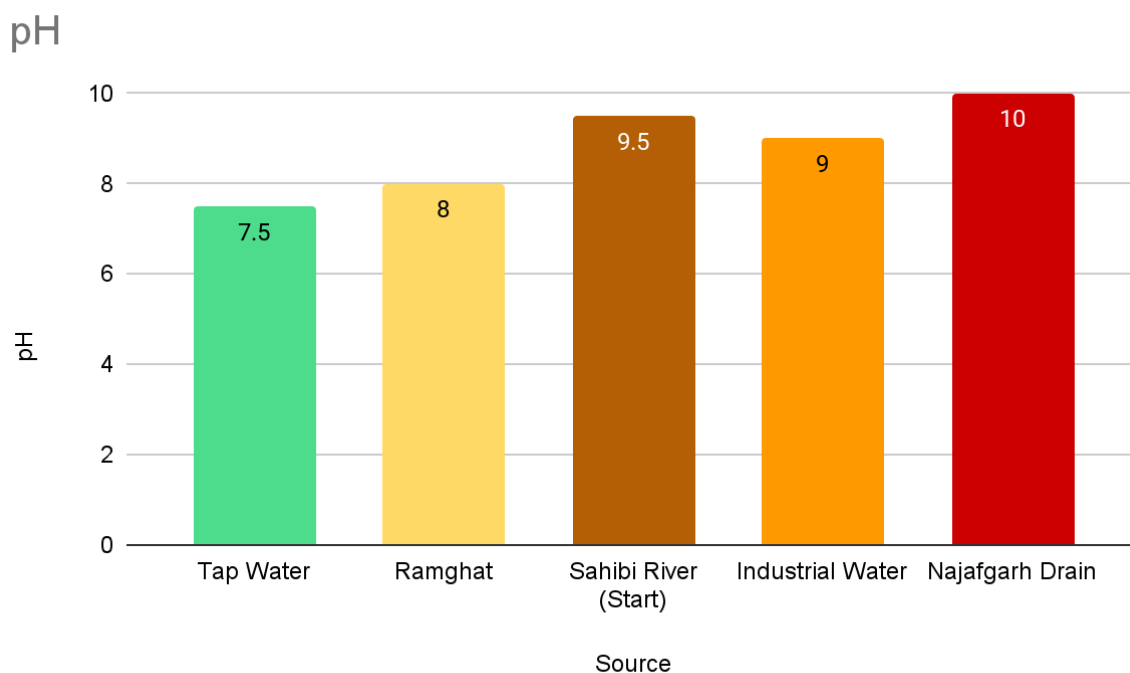
CRITERIA		RATIONALE
1. Fecal Coliform MPN/100 ml:	500 (desirable) 2500 (Maximum Permissible)	To ensure low sewage contamination. Fecal coliform and fecal streptococci are considered as they reflect the bacterial pathogenicity.
2. Fecal Streptococci MPN/100 ml:	100 (desirable) 500 (Maximum Permissible)	The desirable and permissible limits are suggested to allow for fluctuation in environmental conditions such as seasonal change, changes in flow conditions etc.
2. pH:	Between 6.5 –8.5	The range provides protection to the skin and delicate organs like eyes, nose, ears etc. which are directly exposed during outdoor bathing.
3. Dissolved Oxygen:	5 mg/l or more	The minimum dissolved oxygen concentration of 5 mg/l ensures reasonable freedom from oxygen consuming organic pollution immediately upstream which is necessary for preventing production of anaerobic gases (obnoxious gases) from sediment.
4. Biochemical Oxygen demand 3 day, 27°C:	3 mg/l or less	The Biochemical Oxygen Demand of 3 mg/l or less of the water ensures reasonable freedom from oxygen demanding pollutants and prevent production of obnoxious gases”;

Source: https://cpcb.nic.in/wqm/Primary_Water_Quality_Criteria.pdf



1) pH

The values of pH were taken using Litmus Paper Test. The Result can also be seen in Table:4



2) Ammonia Content

The values of Ammonia content were taken by titrating it against standardized HCl solution and using phenolphthalein indicator.

Observation and Calculation of the above test of ammonia:

Normality of standardized HCl = N/20

Volume of each water sample used = 10ml

1) Ramghat

Initial Reading	Final Reading	Volume of HCl used (ml)
0	1.3	1.3
1.3	2.4	1.1
2.4	3.3	0.9

Table 1.1

Volume of HCl used for further calculation = 0.9 ml

Strength of Ammonia = 76.5 ppm



2) Confluence Point

Initial Reading	Final Reading	Volume of HCl used (ml)
0	0.9	0.9
0.9	1.9	1
2.2	3.2	1

Table 1.2

Volume of HCl used for further calculations = 1 ml

Strength of Ammonia = 85 ppm

3) Industrial Waste

Initial Reading	Final Reading	Volume of HCl used (ml)
5.6	6.8	1.2
6.8	7.9	1.1
7.9	8.0	1.1

Table 1.3

Volume of HCl used for further calculations = 1.1 ml

Strength of Ammonia = 93.5 ppm

4) Najafgarh Drain

Initial Reading	Final Reading	Volume of HCl used (ml)
0	4	4
4.2	7.7	3.5
7.7	11.2	3.5

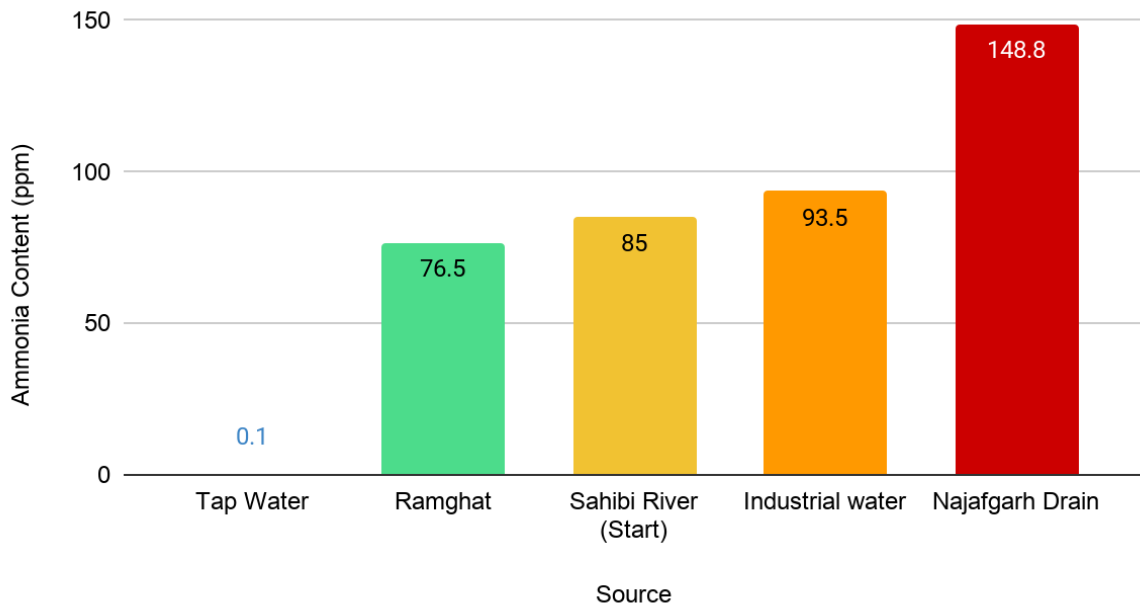
Table 1.4

Volume of HCl used for further calculations = 3.5 ml

Strength of Ammonia = 148.8 ppm



Ammonia Content (ppm) vs. Source



1) Dissolved Carbon Dioxide

1) Ramghat

Initial Reading	Final Reading	Volume of Na ₂ CO ₃ used (ml)
0	5.5	5.5
5.5	11	5.5
11	16.6	5.6

Table 2.1

Volume of Na₂CO₃ used for further calculation = 5.5 ml

Strength of CO₂ = 121 ppm

2) Confluence Point

Initial Reading	Final Reading	Volume of Na ₂ CO ₃ used (ml)
0	6	6
6	12	6
12	18.2	6.2

Table 2.2

Volume of Na₂CO₃ used for further calculations = 6 ml

Strength of CO₂ = 132 ppm



3)Industrial Waste

Initial Reading	Final Reading	Volume of Na ₂ CO ₃ used (ml)
0	20	20
20	40	20
40	61	21

Table 2.3

Volume of Na₂CO₃ used for further calculations = 20 ml

Strength of CO₂ = 440 ppm

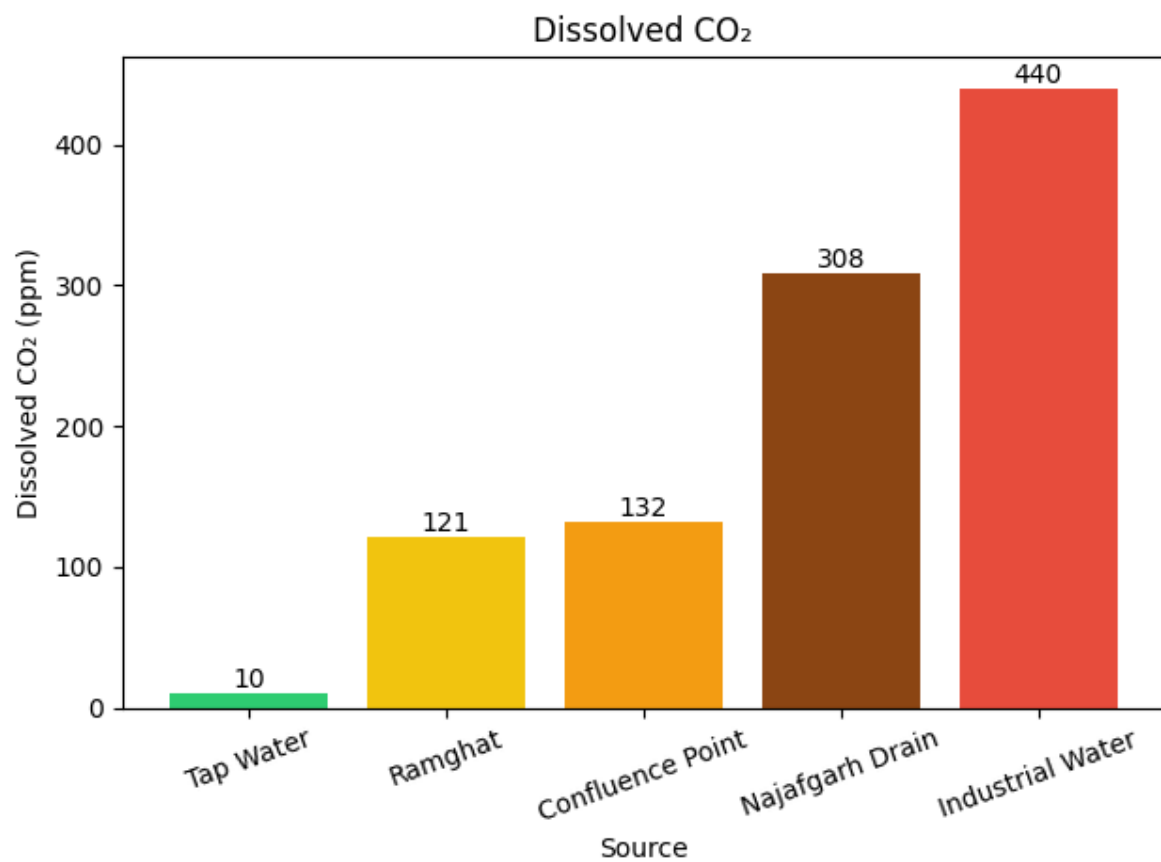
4)Najafgarh Drain

Initial Reading	Final Reading	Volume of Na ₂ CO ₃ used (ml)
0	36	36
36	71	35
71	106	35

Table 2.4

Volume of Na₂CO₃ used for further calculations = 35 ml

Strength of CO₂ = 308 ppm



2) Dissolved Oxygen

1) Ramghat

Initial Reading	Final Reading	Volume of Hypo solution used (ml)
0	0.35	0.35
0.35	0.7	0.35
0.7	1.06	0.36

Table 3.1

Volume of Hypo solution used for further calculation = 0.35 ml

Strength of O₂ = 7 ppm

2) Confluence Point

Initial Reading	Final Reading	Volume of Hypo solution used (ml)
0	0.32	0.32
0.32	0.62	0.3
0.62	0.92	0.3

Table 3.2

Volume of Hypo solution used for further calculations = 0.3 ml

Strength of O₂ = 6 ppm

3) Industrial Waste

Initial Reading	Final Reading	Volume of Hypo solution used (ml)
0	0.1	0.1
0.1	0.2	0.1
0.2	0.31	0.11

Table 3.3

Volume of Hypo solution used for further calculations = 0.1 ml

Strength of O₂ = 2 ppm

4) Najafgarh Drain

Initial Reading	Final Reading	Volume of Hypo solution used (ml)
0	0.25	0.25
0.25	0.5	0.25
0.5	0.77	0.27

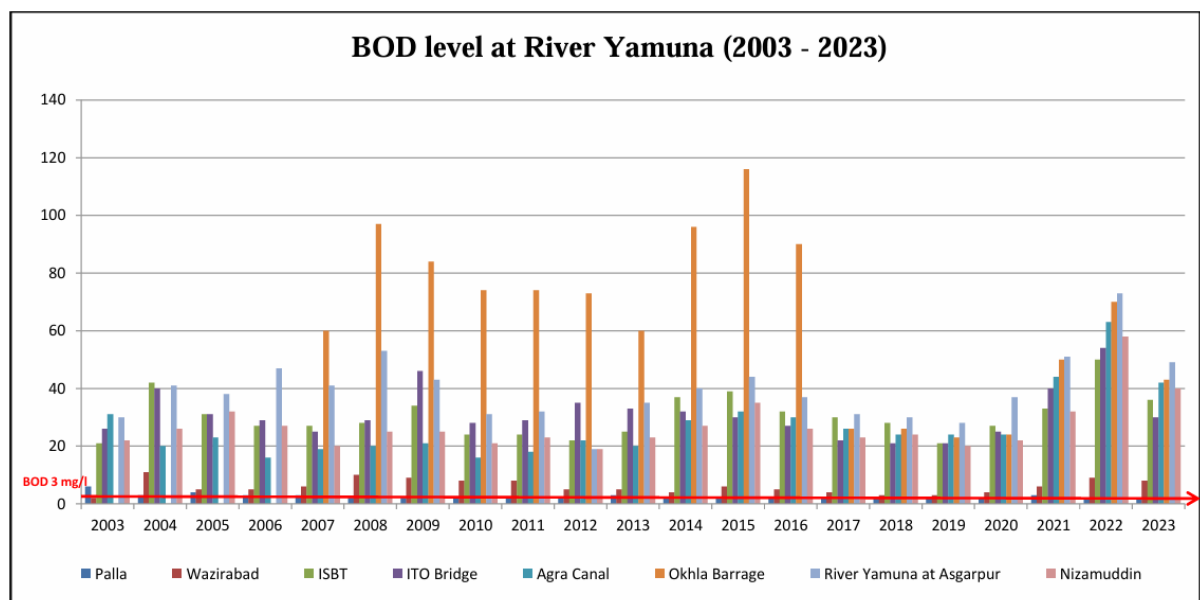
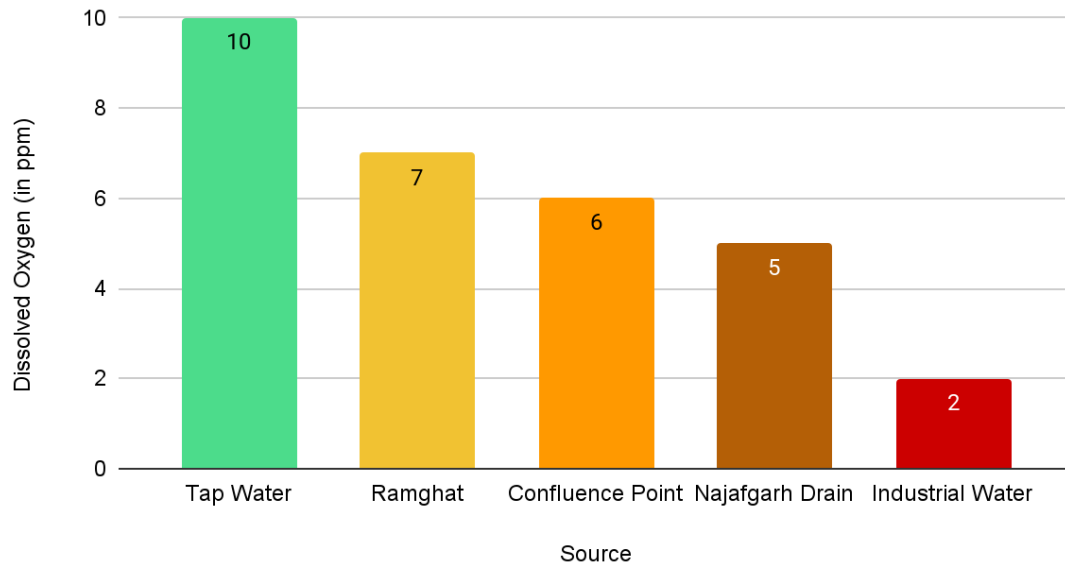
Table 3.4



Volume of Hypo solution used for further calculations = 0.25 ml

Strength of O₂ = 5 ppm

Dissolved Oxygen



Source: <https://dpcc.delhi.gov.in>



Water Quality Parameters (Comparative Data Summary)

The following table summarizes the chemical parameters measured during the study:

Source	pH	Ammonia (ppm)	Dissolved CO2 (ppm)	Dissolved Oxygen (ppm)
Tap Water (Control)	7.5	0.1	10	10
Ramghat	8.0	76.5	60	7
Confluence Point	9.5	85.0	70	6
Industrial Water	9.0	93.5	190	2
Najafgarh Drain	10.0	148.8	308	5

Table: 4

Environmental and Ecological Impact

- **Biodiversity Collapse:** Near-zero dissolved oxygen levels in critical stretches have created hypoxic "dead zones" where sensitive species cannot survive.
- **Mass Mortality:** Severe oxygen depletion causes widespread mortality among fish and other aquatic organisms.
- **Toxicity and Bioaccumulation:** Industrial discharge introduces heavy metals and organic pollutants that concentrate in the food chain, posing risks to wildlife and human consumers.

Experimental Limitation

Dissolved CO₂ values were calculated using different sample volumes (20 mL for river samples and 50 mL for the Najafgarh drain sample) based on experimental feasibility. Since dissolved gases are sensitive to environmental conditions, factors such as sample storage duration (2–3 days), temperature variation, and exposure to air may have influenced the observed titration readings. Therefore, the calculated CO₂ values represent relative pollution intensity and trends rather than exact in-situ equilibrium concentrations.



REPORTS FROM GOVERNMENT BODIES

WATER QUALITY OF RIVER YAMUNA IN DELHI (January, 2021)

STN Code	Name Of Monitoring Location	Dissolved Oxygen (mg/L)	pH	Conductivity (µmhos/cm)	BOD (mg/L)	Fecal Coliform (MPN/100ml)	Total Coliform (MPN/100ml)
WATER QUALITY CRITERIA		> 5 mg/l	6.5-8.5	–	< 3 mg/l	< 2500 MPN/100ml	–
1120	PALLA	9.6	7.8	689	6.2	22000	54000
1121	NIZAMUDDIN	0.3 (BDL)	7.4	1488	29	790000	1100000
1375	OKHLA BRIDGE (INLET OF AGRA CANAL)	0.3 (BDL)	7.4	1001	27	700000	1100000
1812	AT OKHLA AFTER MEETING OF SHAHDARA DRAIN	0.3 (BDL)	7.5	1225	57	3300000	3300000
2057	AGRA CANAL, MADANPUR KHADAR	0.3 (BDL)	7.4	1015	37	790000	2400000

BDL- Below Detection Limit (0.3 mg/L)

WATER QUALITY OF RIVER YAMUNA IN DELHI (February, 2021)

STN Code	Name Of Monitoring Location	Dissolved Oxygen (mg/L)	pH	Conductivity (µmhos/cm)	BOD (mg/L)	Fecal Coliform (MPN/100ml)	Total Coliform (MPN/100ml)
WATER QUALITY CRITERIA		> 5 mg/l	6.5-8.5	–	< 3 mg/l	< 2500 MPN/100ml	–
1120	PALLA	13.5	7.9	1180	7	68	1200
1121	NIZAMUDDIN	0.3 (BDL)	7.5	1892	34	4900000	8400000
1375	OKHLA BRIDGE (INLET OF AGRA CANAL)	0.3 (BDL)	7.4	1342	32	380000	940000
1812	AT OKHLA AFTER MEETING OF SHAHDARA DRAIN	0.3 (BDL)	7.6	1792	83	2100000	14000000
2057	AGRA CANAL, MADANPUR KHADAR	0.3 (BDL)	7.4	1348	32	140000	790000

BDL- Below Detection Limit (0.3 mg/L)

Source: https://cpcb.nic.in/NGTMC/WQD_jan-feb_2021.pdf

Station Code	Monitoring Location	State Name	Temperature (°C)		Dissolved Oxygen (mg/L)		pH		Conductivity (µmho/cm)		BOD (mg/L)		Nitrate N (mg/L)		Fecal Coliform (MPN/100ml)		Total Coliform (MPN/100ml)		Fecal Streptococci (MPN/100ml)	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Primary Water Quality Criteria notified under The E (P) Rules, 1986					> 5.0 mg/L		6.5 - 8.5				< 3.0 mg/L				< 2500 MPN/ 100 MI				< 500 MPN/ 100 MI	
5098	RIVER YAMUNA AT WAZIRABAD	DELHI			4.0	8.2	7.2	7.5			4.0	9.8	0.00	0.00	200	5500				
5099	RIVER YAMUNA AT ISBT BRIDGE	DELHI			0.3	3.0	7.1	7.5			22.0	46.0	0.00	0.00	5500	100000	17000	17000		
5100	RIVER YAMUNA AT ITO BRIDGE	DELHI			0.3	4.1	7.2	7.5			18.0	38.0	0.00	0.00	6100	81000				
1121	RIVER YAMUNA AT NIZAMUDDIN, DELHI	DELHI			0.3	2.5	7.1	7.5			26.0	48.0	0.00	0.00	9300	31000				
1375	RIVER YAMUNA AT OKHLA BRIDGE (INLET OF AGRA CANAL), DELHI	DELHI			0.3	1.8	7.1	8.0			30.0	50.0	0.00	0.00	120000	380000				
1812	RIVER YAMUNA AT OKHLA AFTER MEETING OF SHAHDARA DRAIN, DELHI	DELHI			0.3	1.1	6.8	7.7			35.0	56.0	0.00	0.00	170000	470000				

Source: https://cpcb.nic.in/wqm/2023/WQuality_River-Data-2023.pdf



Proposed Solutions and Recommendations

A multi-pronged approach is required for river restoration:

- **Wastewater Management:** Upgrade Sewage Treatment Plants (STPs) to achieve 100% treatment and doing as less corruption as possible (best case scenario: **no corruption**).
 - **Industrial Regulation:** Enforce strict effluent standards and "Zero Liquid Discharge" (ZLD) policies.
 - **Monitoring:** Implement real-time sensors and a central data dashboard.
 - **Public Awareness:** Organize community clean-up drives and reduce direct waste dumping.
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Conclusion

The study clearly demonstrates a significant deterioration in the water quality of the Yamuna River as it flows through Delhi. The experimental results show a marked increase in ammonia and dissolved CO₂ levels, along with a decrease in dissolved oxygen, indicating rising **organic pollution** and **microbial activity**.

The confluence point shows noticeable impact due to mixing with the Najafgarh Drain, while the industrial region exhibits extreme chemical imbalance, with very high CO₂ levels and critically low dissolved oxygen. These conditions indicate the development of biologically stressed or **near-dead zones** in certain stretches of the river.

The findings confirm that sewage discharge, industrial effluents, and drain inputs are the primary contributors to pollution, leading to severe ecological degradation and reduced aquatic life sustainability.

Therefore, restoration of the Yamuna is not only an environmental necessity but also a critical public health priority, requiring strict regulation, improved wastewater treatment, and continuous monitoring

END OF REPORT



GALLERY



